CHAPTER

ATA 72

ENGINE

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Contents

Engine – General

Introduction

The purpose of the engine is to provide motive power to the aircraft and to supply power to various aircraft systems.

General Description

The PW150A turboprop engine has an axial, Low Pressure (LP) compressor and a centrifugal, High Pressure (HP) compressor. The engine also has a reverse flow annular combustor and a two−stage power turbine. The power turbine provides the drive for the compressors and the Reduction Gearbox (RGB). The RGB reduces the power turbine input speed and increases the torque.

Air enters the compressors through the inlet case. The compressors are driven by the LP and HP turbines. Fuel is added to the compressed air in the combustion chamber. The igniters are used for starting but are not required once the fuel/air mixture has ignited.

Detailed Description

Refer Figure 72-1

The engine is started by the starter/generator rotating the HP impeller and the turbine. The drive for this rotation is taken through the accessory gearbox and down the associated driveshaft to rotate the bevel gear splined to the HP impeller.

Fuel is sprayed into the combustion chamber, where it is mixed with the incoming air from the compressors. At a predetermined speed, the ignition system switches the igniters on and the air/fuel mixture is ignited. The resultant rearward flow of expanding gases drives the HP and LP turbines, which are connected to the HP and LP compressors.

The rotating turbines turn the compressors which draw in more air to mix and burn with the fuel. This creates additional expanding gases which increase the speed of the turbines and compressors until the engine has achieved self−sustaining speed with continuous combustion. The igniters are then switched off, and the starter/generator functions as a generator.

The flow of expanding gasses through the engine also drives the power turbine, which turns the propeller through the gearbox. A further increase in fuel flow to the combustion chamber will increase gas expansion, resulting in an increase in turbine and compressor speed.

The HP and power−turbine rotors rotate clockwise, and the LP rotor rotates counterclockwise when viewed from the rear.

The engine has two modules, they are:

- The reduction gearbox module
- The turbomachinery module

The turbomachinery module consists of the following sub-modules:

- Air Inlet
- Compressor Section
- Combustion Section
- Power Turbines

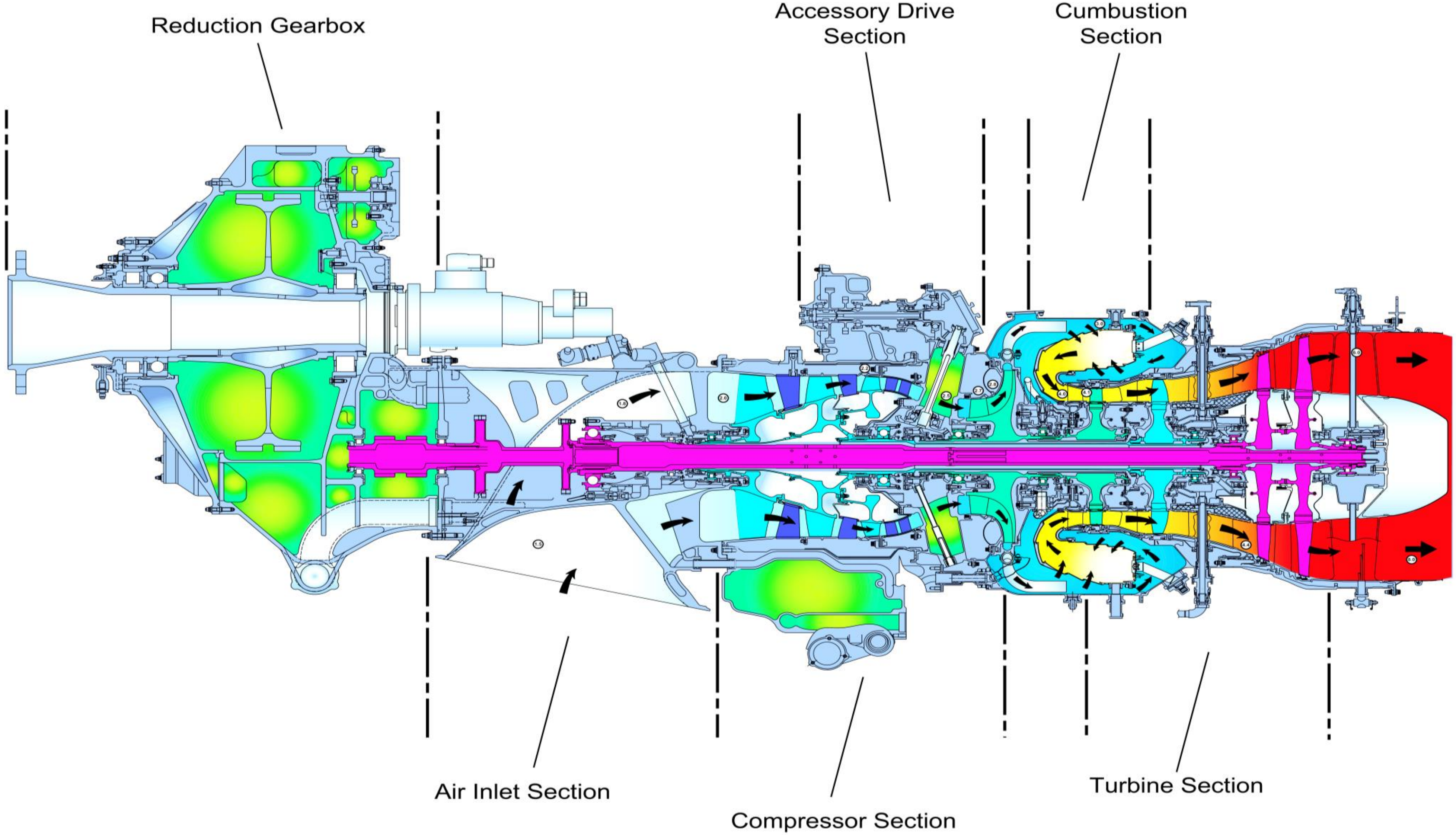


Figure 72-1 Engine Cutaway View

Reduction Gear Box (RGB)

Introduction

The Reduction Gearbox reduces the input speed from the engine, to a suitable speed range for use by the propeller.

General Description

Refer Figure 72-2 & *72-3*

The RGB reduces the power turbine shaft speed and increases the torque. The output from the RGB is used to drive the propeller. The RGB has two housings, the front housing and the rear housing.

These housings contain a double−reduction gear−train. The first stage reduction gear−train gets power from the power turbine shaft which is connected to the RGB input shaft.

Accessory gears drive the ac generator, hydraulic pump and propeller overspeed governor which are mounted on the accessory drive cover. The propeller control unit is mounted below the accessory drive cover and is connected to the rear of the propeller shaft. An electrically driven feathering pump is also mounted on the rear of the RGB.

Engine mounting pads are located on each side and on the top of the rear housing.

Lifting brackets are located at the one and eleven o'clock positions on the front housing flange.

The Reduction Gearbox (RGB) module is one of the two primary modules of the engine. The RGB is the reduction stage between the power turbine shaft (input) and the propeller shaft (output). The total reduction ratio is 17.16 to 1.

The RGB reduces the power turbine speed (17,501 rpm) to one suitable for propeller operation (1020 rpm) and has the functions that follow:

- Support the propeller
- Transmit turbomachinery power to propeller
- Transmit thrust to aircraft structure
- Drive the RGB mounted accessories, a/c generator, overspeed governor and pump, and hydraulic pump
- Give appropriate reference to torque indicating system.

The first stage of the reduction geartrain has a helical input shaft and two first−stage helical gears. The second stage has two spur gearshafts (pinion gears) and one large spur gearshaft (bull gear) installed on the propeller shaft.

The first−stage gears are connected to the second−stage gearshafts (pinion gears) through the flexible layshaft couplings. The first−stage gears, the flexible layshaft couplings and the second−stage gearshafts make up the flexible layshafts.

The four accessory gears in the reduction gearbox are as follows:

- Idler−drive spur gearshaft
- Alternator−drive spur gearshaft
- Hydraulic pump drive spur gearshaft
- Overspeed−governor drive spur gearshaft

The reduction gearbox assembly includes three housings: the front housing, the rear housing, and the diaphragm input−drive housing.

The accessory drive cover is installed on the top of the back face of the rear housing. The cover has three mounting pads that follow:

- Alternator
- Hydraulic pump
- Overspeed−governor

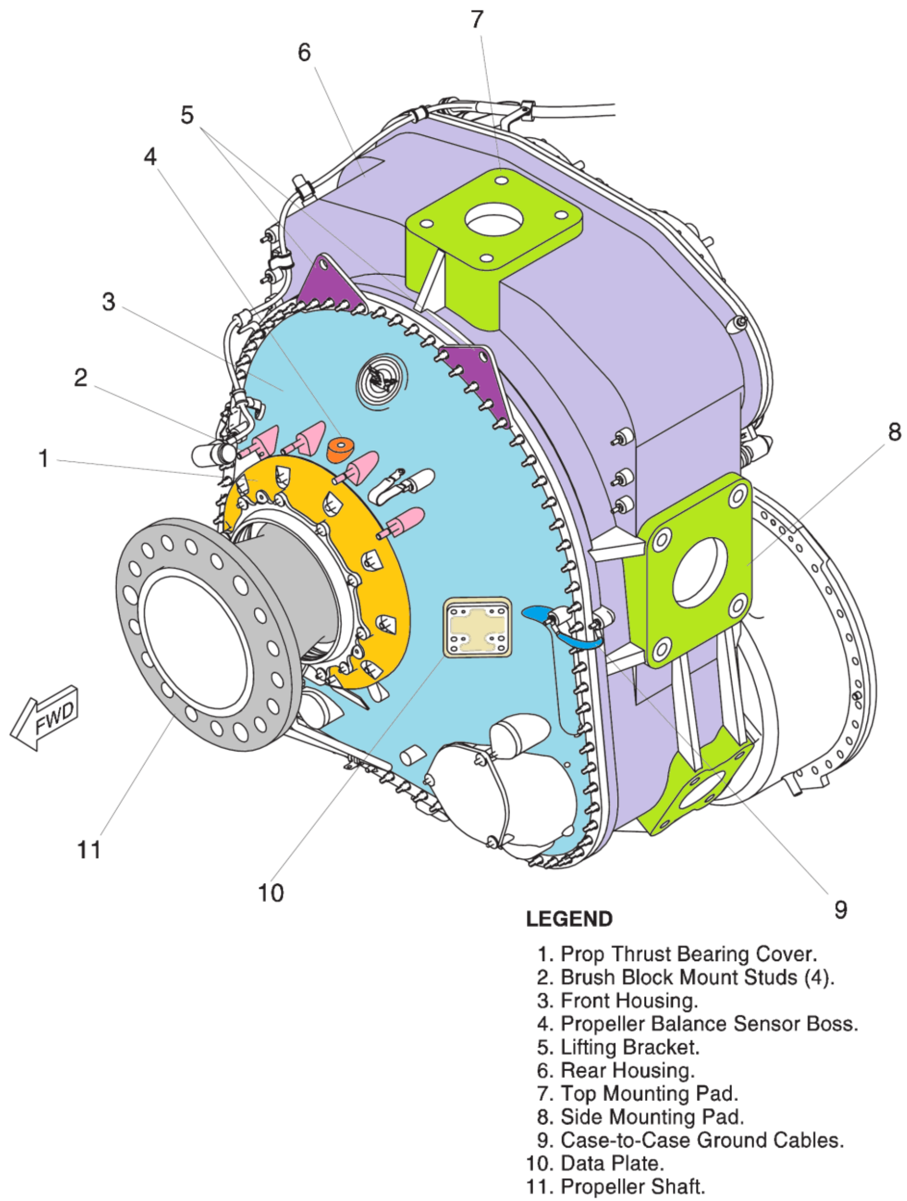


Figure 72-2 Reduction Gear Box (RGB) – Front Detail

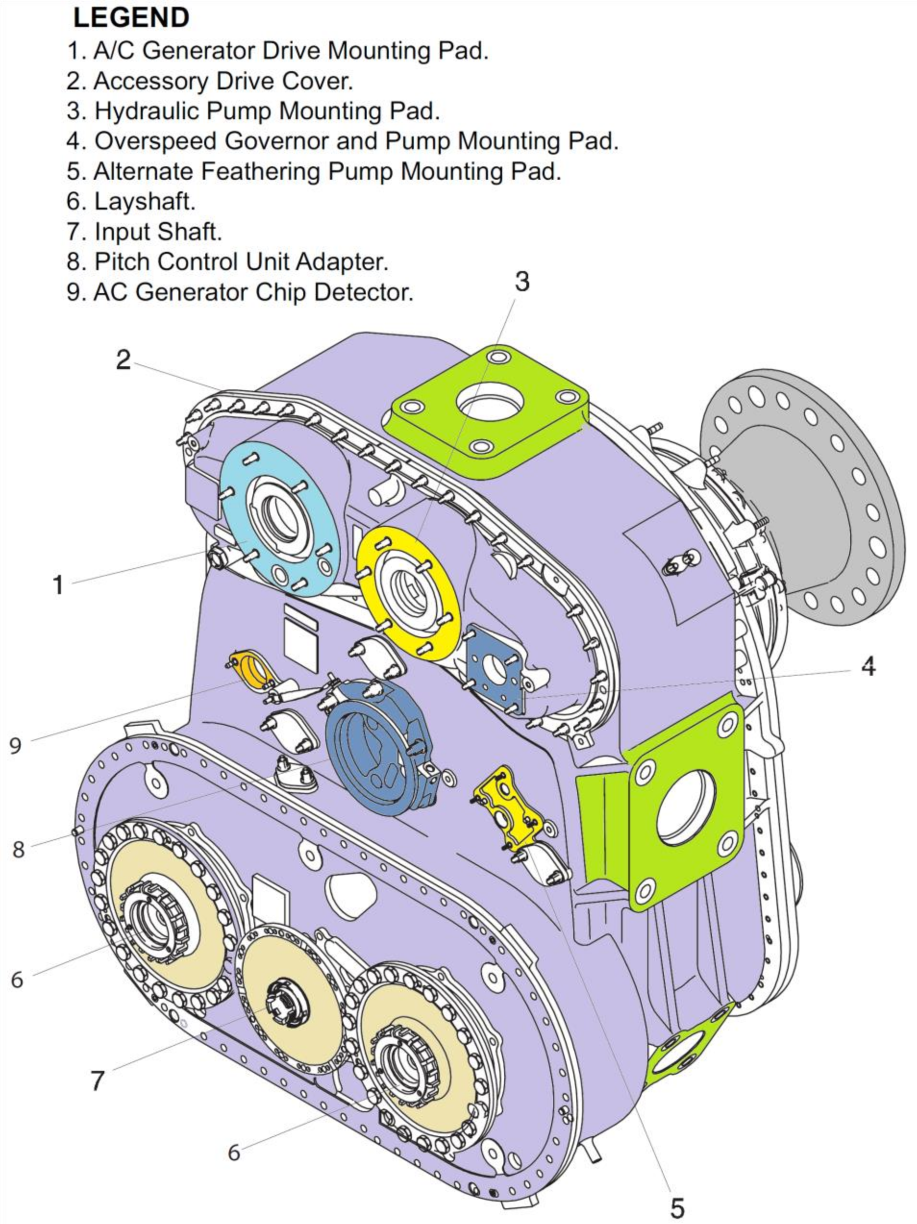


Figure 72-3 Reduction Gear Box (RGB) - Rear Detail

Detailed Description

Refer Figure 72-4

When the engine turns, the power turbines drive the input shaft clockwise through the PT shaft and the torque shaft. The input shaft engages the first−stage gears. The first−stage gears turn counterclockwise and the flexible layshaft couplings transmit the power to the second−stage gearshafts (pinion gears). The first−stage reduction ratio is 3.92 to 1.

The two second−stage gearshafts (pinion gears) drive the second−stage spur gearshaft (bull gear) clockwise, together with the propeller shaft. The second−stage reduction ratio is 4.37 to 1.

The bull gear drives the idler−drive gearshaft and the overspeed−governor gearshaft counterclockwise. The idler−drive gearshaft drives the alternator−drive gearshaft clockwise. The reduction ratios from the bull gear to the accessory gearshafts are as follows:

- The idler−drive gearshaft: 5.62 to 1
- The alternator−drive gearshaft: 15.79 to 1
- The overspeed−governor gearshaft: 5.62 to 1
- The hydraulic pump gearshaft: 6.88 to 1.

A chamber in the rear housing is supplied with engine oil. This auxiliary tank is always full of oil, even when the engine is not in operation. Oil from this tank lubricates the reduction and accessory gears and bearings through oil passages, oil jets and nozzles. The overspeed governor and the propeller control unit use this oil to operate the pitch−change mechanism of the propeller. The electric feathering−pump mounting pad on the rear housing has oil ports that are connected to this auxiliary oil tank.

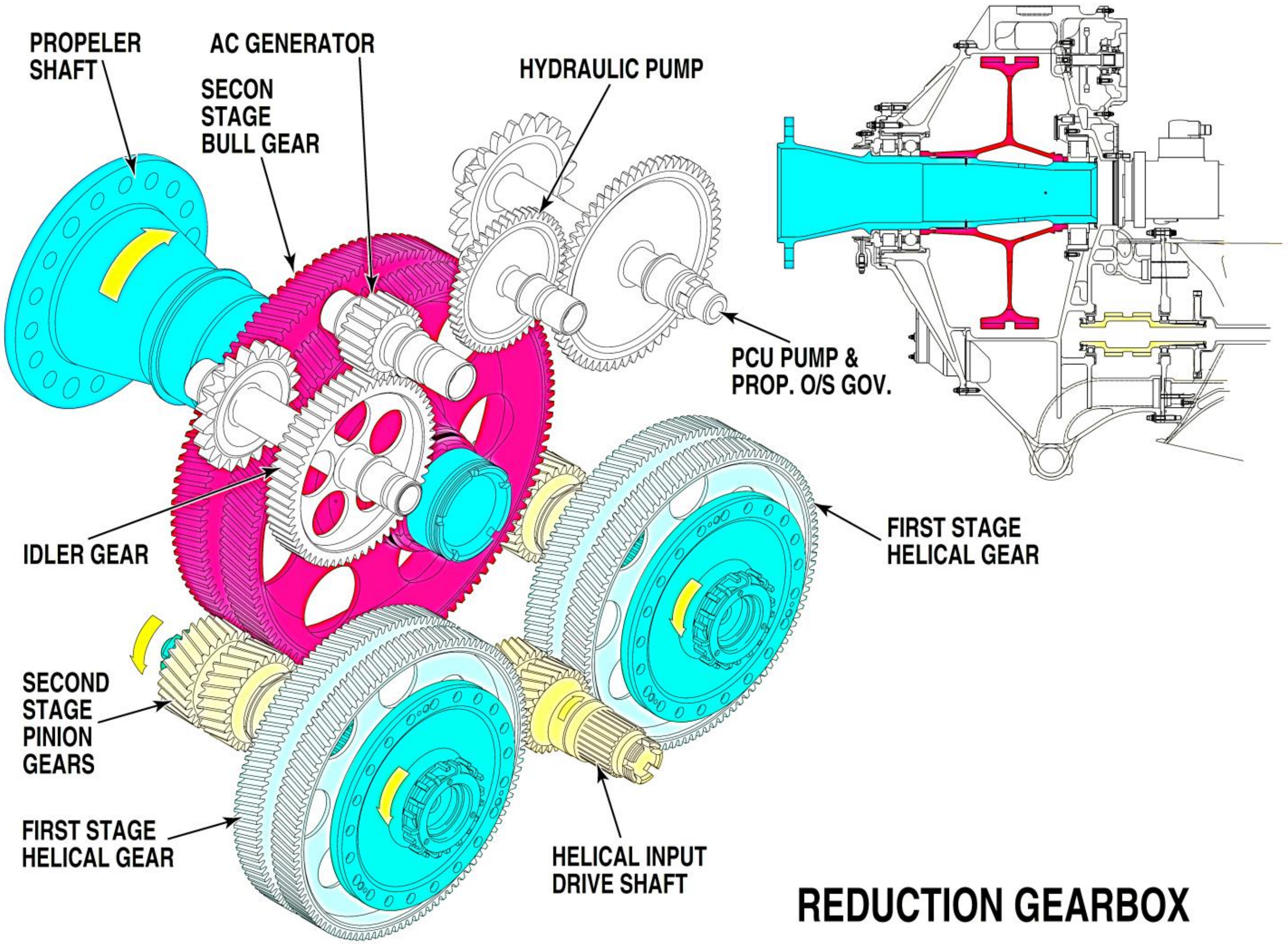


Figure 72-4 RGB Gear Trains

Air Intake

Introduction

The inlet case provides air to the Low Pressure (LP) compressor under all flight conditions with as little turbulence and pressure variation as possible.

General Description

The function of the air intake is performed by the front inlet case. The front inlet case is the entrance for the air used by the engine.

Front Inlet Case

Refer Figure 72-5

The air inlet is located between the Reduction Gearbox (RGB) and the Low Pressure (LP) compressor case. The front inlet case is made of magnesium alloy. Hot oil flows through internal passages in the case. This prevents the formation of ice on the inlet lip.

The purpose of the front inlet case is to:

- Provide structural connection between the RGB and the turbomachinery
- Direct air flow properly between the intake housing and the first stage Integrated Blade Rotor (IBR)
- Support power turbine and low pressure shafts
- Support the flexible coupling shaft assembly that transmits power to the RGB
- Provide sealing for the no.1 bearing cavity
- Allow oil to pass to bearing cavities.

The Full Authority Digital Electronic Control (FADEC) is mounted on the left side of the case and the fuel flowmeter is mounted on the right side.

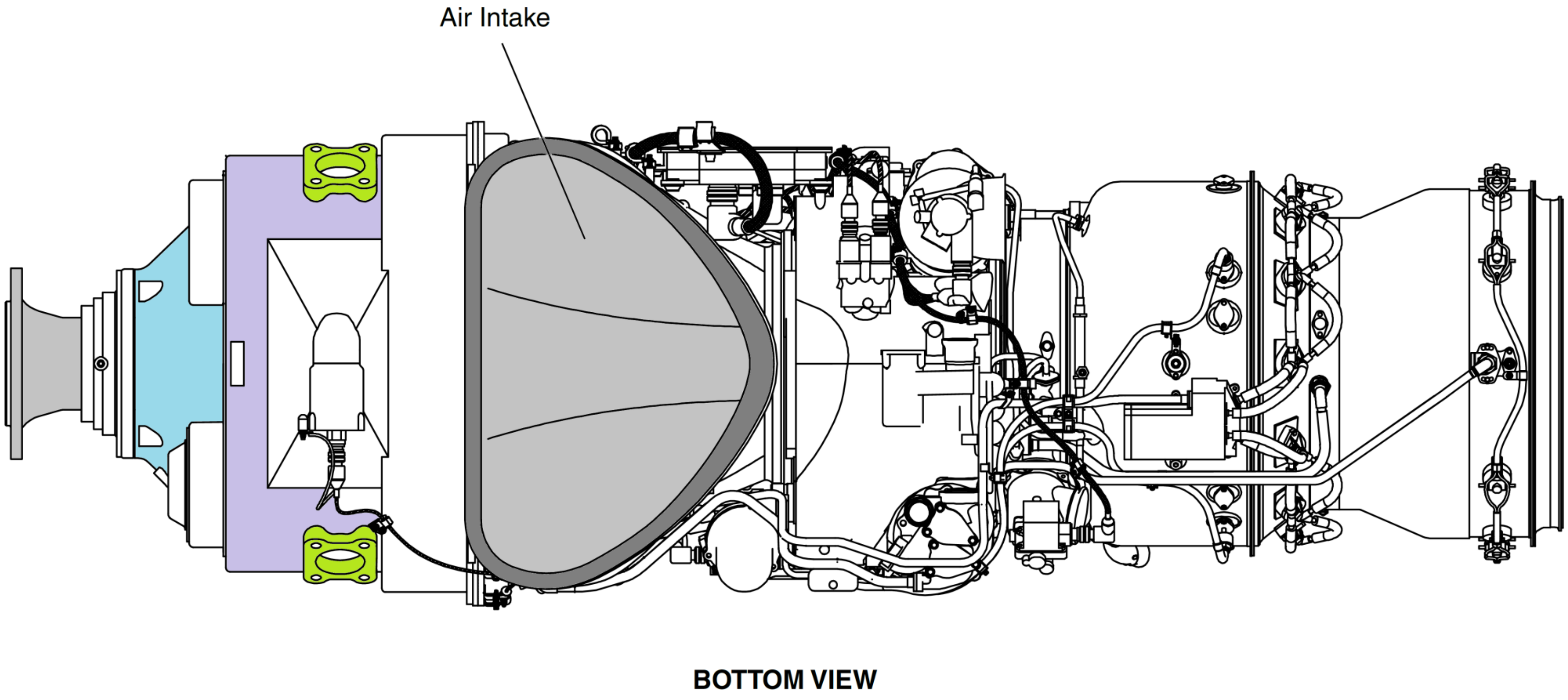


Figure 72-5 Engine Air Intake

Compressor Section

Introduction

The compressor raises the pressure of the incoming air before passing it to the combustion chamber.

General Description

The compressors draw air into the engine through the front inlet case. Air pressure increases across an axial (three stages) compressor and a centrifugal impeller. The compressed air is then delivered to the combustion chamber and other systems.

The compressed air is used to:

- Sustain combustion to give the energy necessary to rotate the compressors and power (propeller) sections
- Cool the main hot section components
- Supply bleed air for the aircraft environmental control system (air conditioning and de−icing)
- Seal bearing cavities
- Assist oil scavenging of some bearing cavities
- Operate the two bleed valves.

Detailed Description

Refer Figures 72-6A, 72-76B & 72-7

The compressor section has two independent compressors: an axial three stage compressor named Low Pressure (LP) Compressor, and a centrifugal impeller named High Pressure (HP) Compressor. Each compressor has its own single−stage turbine to make it turn. The axial and centrifugal compressors and the power turbines are attached to co−axially located shafts which are supported by roller and ball bearings.

The gas path is based on a 18 to 1 compression ratio. Carbon seals are used in the compressor and other areas of the turbomachinery for sealing the bearing cavities. The Carbon element is replaceable to reduce maintenance costs.

Air enters the compressor section from the front inlet case. At the first three stages of compression (LP compression), a series of axial rotors and vane cascades compress, direct, and diffuse the air. The fourth stage of compression (HP compression) comes from the HP impeller, from which the air goes through the diffuser pipes.

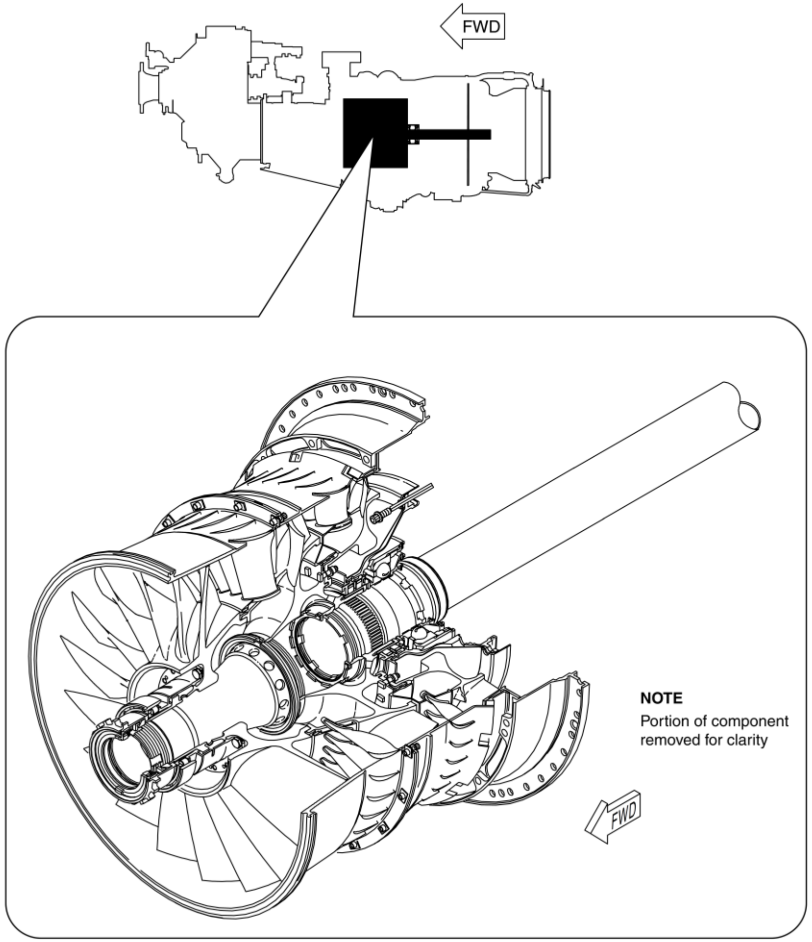


Figure 72-6A Compressor Section-1

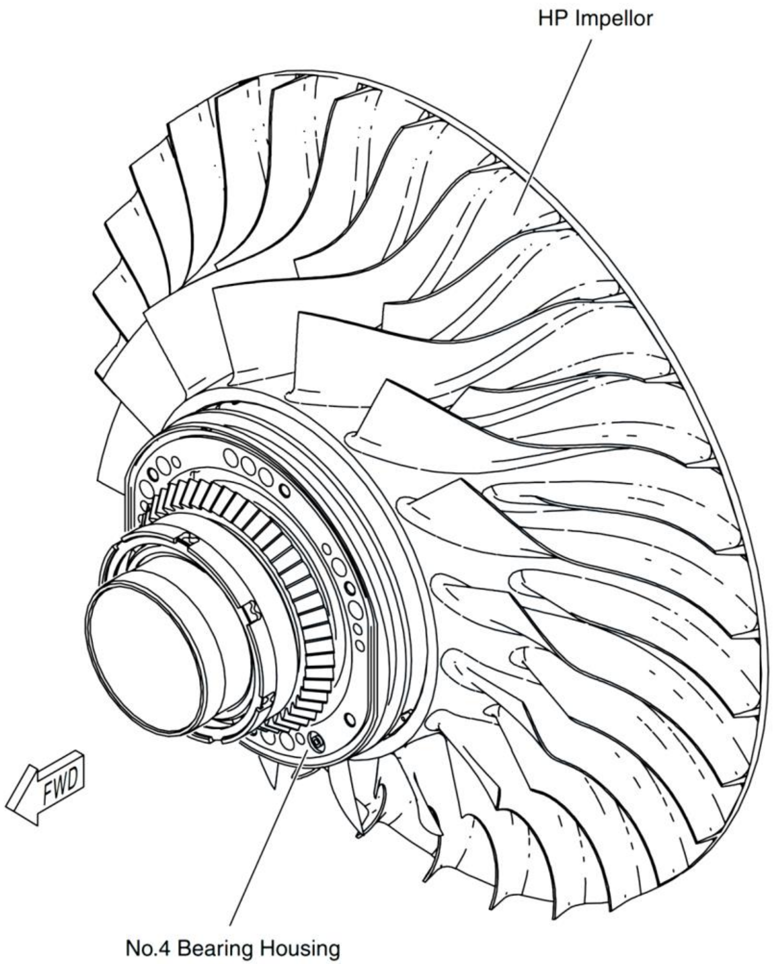


Figure 72-6B Compressor Section-2

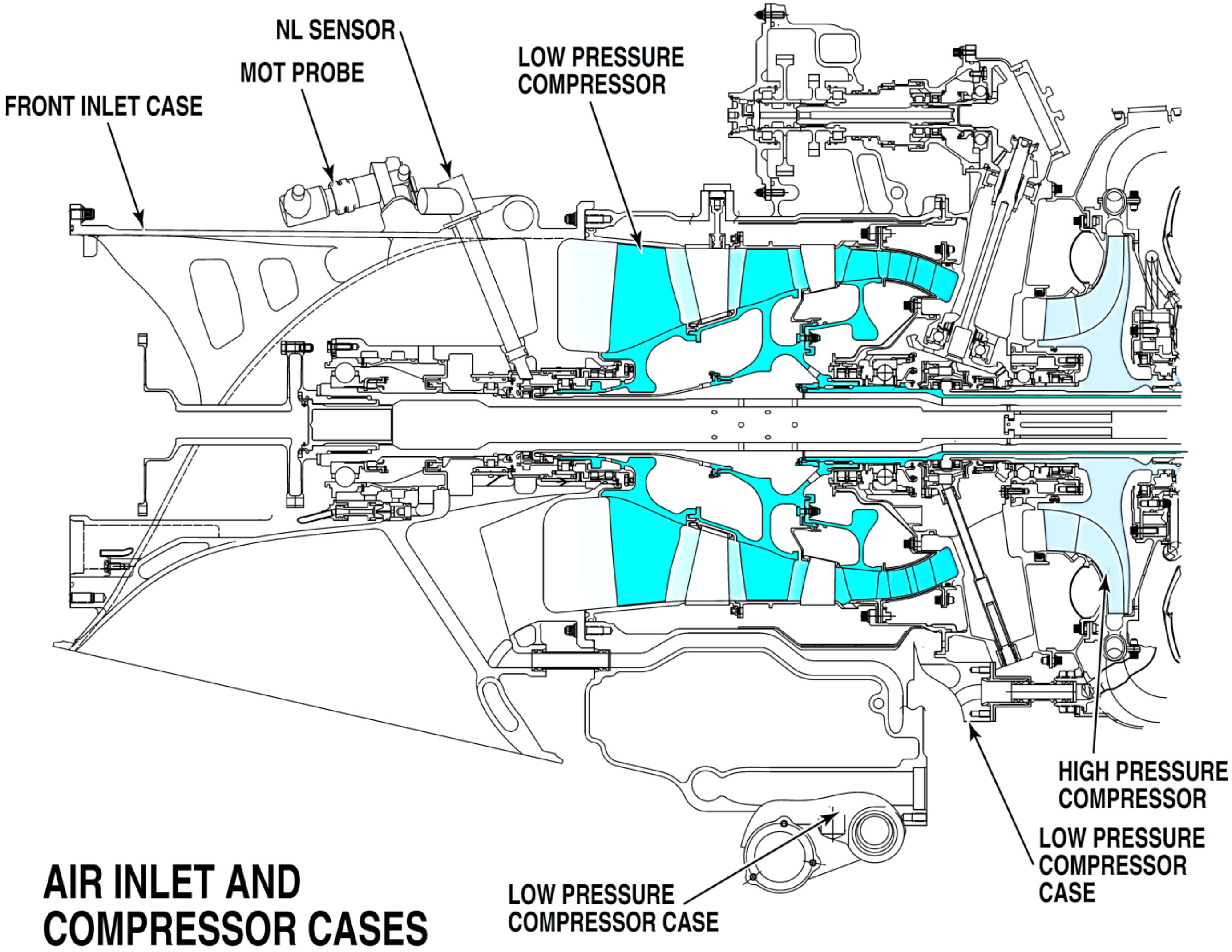


Figure 72-7 Compressor Locator

Compressor Section – Component Description

Low Pressure (LP) Compressor Case

Refer Figure 72-7

The LP compressor case is located between the inlet case and the intercompressor case. The LP compressor case has an integral oil tank at the bottom, and an Accessory Gearbox (AGB) with the deaerator on top. The AGB has two accessory mounting pads. One pad is for the DC starter/generator and the other is for the Permanent Magnet Alternator (PMA) and the Fuel Metering Unit (FMU). The case also contains the no.3 bearing that supports the low pressure shaft. There are two borescope ports access to the 2nd and 3rd stage compressor. The case supplies installation for the components that follow:

- Two Nh speed sensors
- Oil level sight glass and filler neck
- Oil pressure regulating valve
- Main oil filter housing
- Ignition Exciter Box
- Fuel Heater
- Turbomachinery chip detector
- Oil Pump pack
- Deaerator
- Retimet breather (in AGB)
- P2.2 Intercompressor Bleed Valve

The magnesium case offers some containment capability, through the erosion/thermal shield mounted in the case. The principal containment for the LP compressor blades is by the bolted vane and shroud arrangements. An erosion shield is installed in the case to protect the magnesium from hot P2.2 air and debris.

Low Pressure Compressor

Refer Figure 72-7 & 72-8

The LP compressor is supported by the damped No. 2.5 roller bearing and the No.3 ball bearing. The rotor components are three axial Integrated Blade Rotors (IBR) (1st, 2nd, and 3rd stage) all of titanium alloy. The IBRs are designed to be FOD resistant.

The drive comes from the 2nd stage IBR integral stub shaft. The shaft rotates the 3rd stage IBR bolted to a flange next to the 2nd stage IBR. The 1st stage IBR is axially retained to the 2nd stage IBR through a tie shaft arrangement.

The 1st stage vane assembly is a full vane ring cascade. The assembly has an integrated (pinned in) Vane Passage Casing Treatment (VPCT) which increases stall margin. The 2nd stage vane assembly is identical to the 1st stage with the exception that the VPCT is assembled at engine assembly level. Both the 1st and 2nd stage stators are damped. Facing the 3rd stage stator are the bleed slots (P2.2), which bleed up to 15% core flow on the running line. The 3rd stage stator assembly is a non-damped double row cascade. This double row cascade slows the air down to provide near zero swirl at the inlet to the Intercompressor Case (ICC) duct. This assembly is split in two halves to permit one piece compressor module assembly and removal.

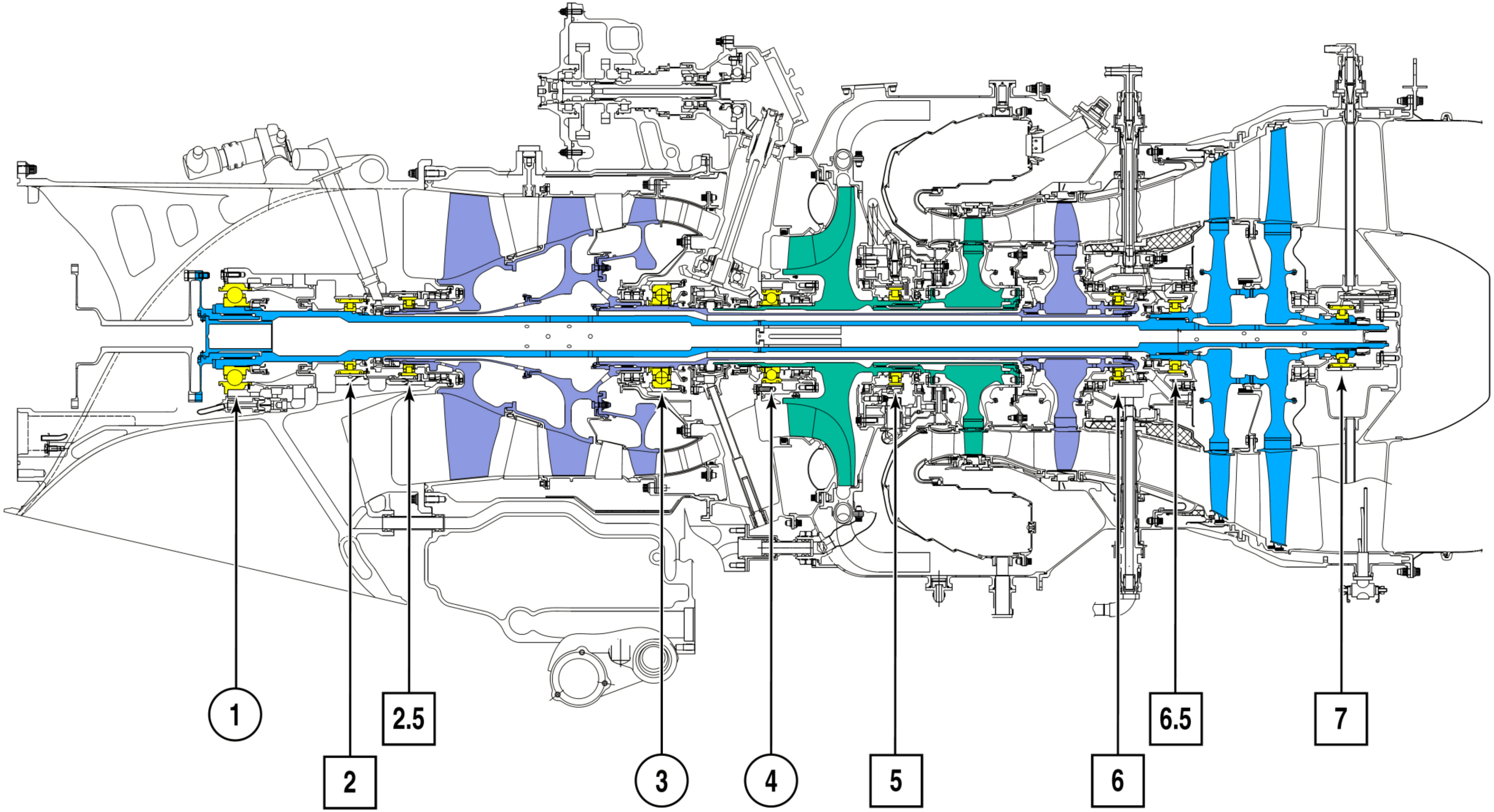


Figure 72-8 Engine Bearings

Inter Compressor Case

Refer Figure 72-10

The intercompressor case is located between the LP compressor case and the gas generator case. It houses the centrifugal HP compressor and the accessory drive shaft (tower shaft).

The intercompressor case forms a path that allows air to flow from the LP compressor to the HP compressor. The titanium intercompressor case casting holds the no. 3 and no. 4 bearing cavity that support the LP and HP compressors. The intercompressor case gives support for the components that follow:

- P2.7 check valve (for ECS)
- P2.7 Handling bleed valve (HBV)
- Angle drive gearbox
- Rear engine mounts
- Drain valve
- HP Compressor

HP Compressor

Refer Figure 72-9

The centrifugal compressor is supported on an integral shaft. The shaft is supported by the upstream side of No. 4 ball bearing and the downstream side of the damped No. 5 roller bearing. The impeller shroud is a double annulus for the P2.7, and P2.8 bleeds. Cabin bleed is provided by P3 and P2.7 air.

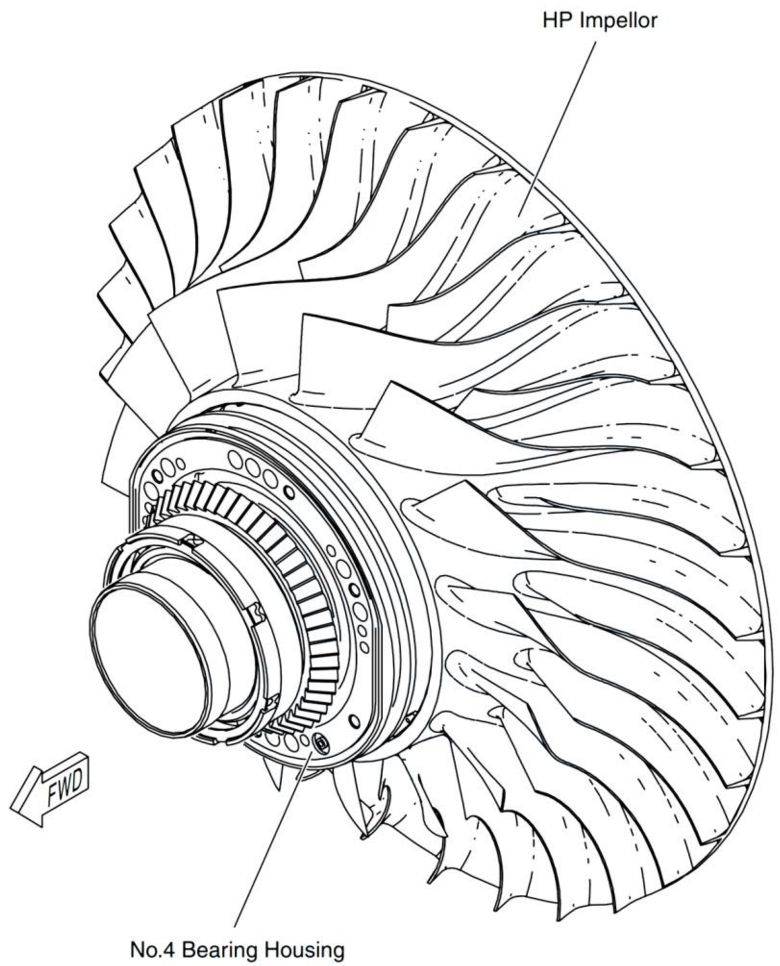


Figure 72-9 HP Compressor/Impeller

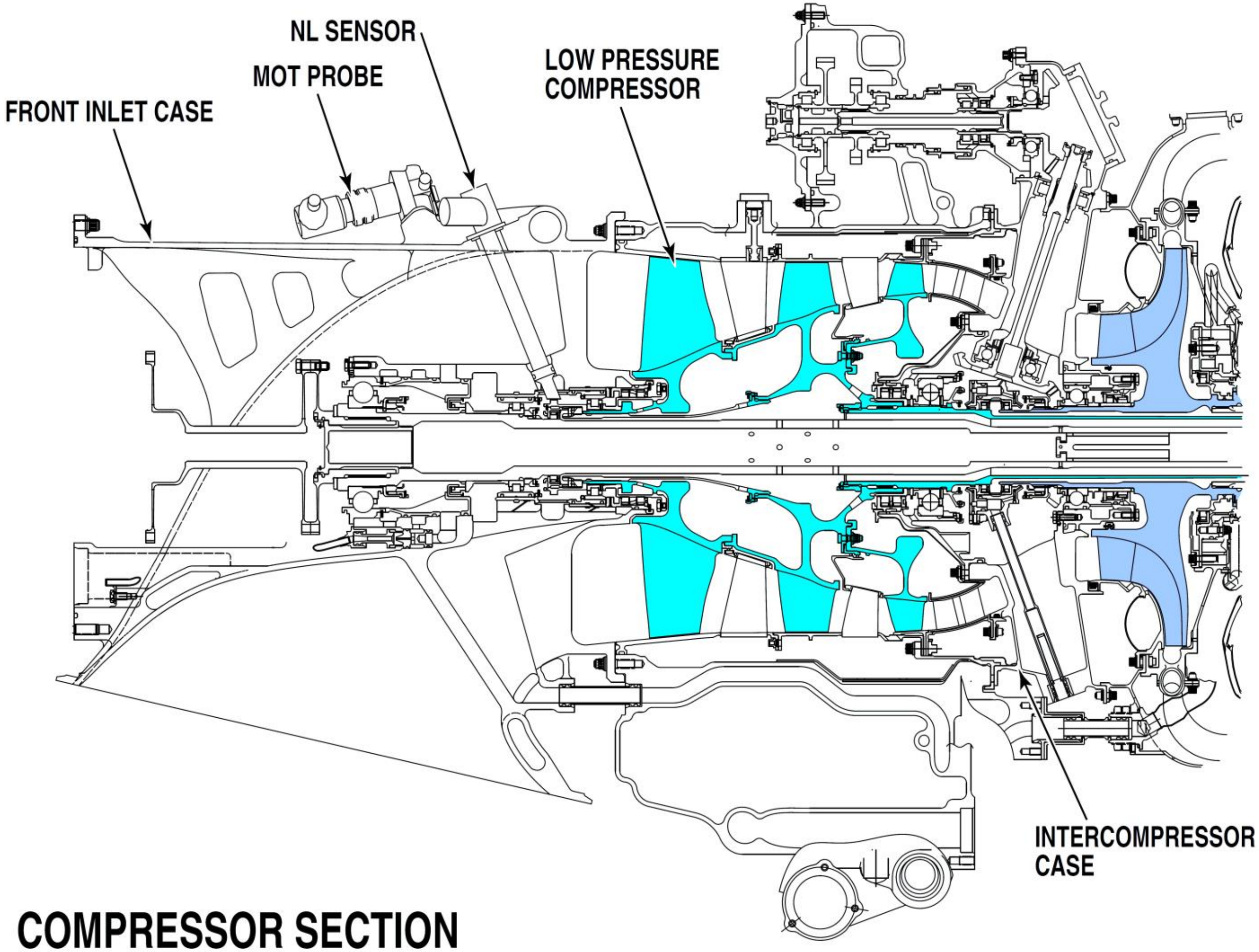


Figure 72-10 Compressor Locator – 2

Combustion Section

Introduction

The combustion section provides an area for the combustion of the fuel/air mixture.

General Description

Fuel is added to the compressed air in the combustion chamber. The igniters are used for starting but are not required once the fuel/air mixture is lit.

Detailed Description

Refer Figure 72-11

The annular reverse−flow combustion chamber is contained in the gas generator case. Fuel manifolds, mounted on the exterior of the turbine support case, supply fuel to nozzles. The nozzles are mounted on the turbine support case and protrude into the combustion chamber liner from the rear. Two igniter plugs are provided on the gas generator case with corresponding bosses in the combustion chamber liner. The gas generator case incorporates an air bleed pad through which P3 air is supplied to the Environmental Control System.

The gas generator case incorporates an air bleed pad through which P3 air is supplied to the Environmental Control System (ECS).

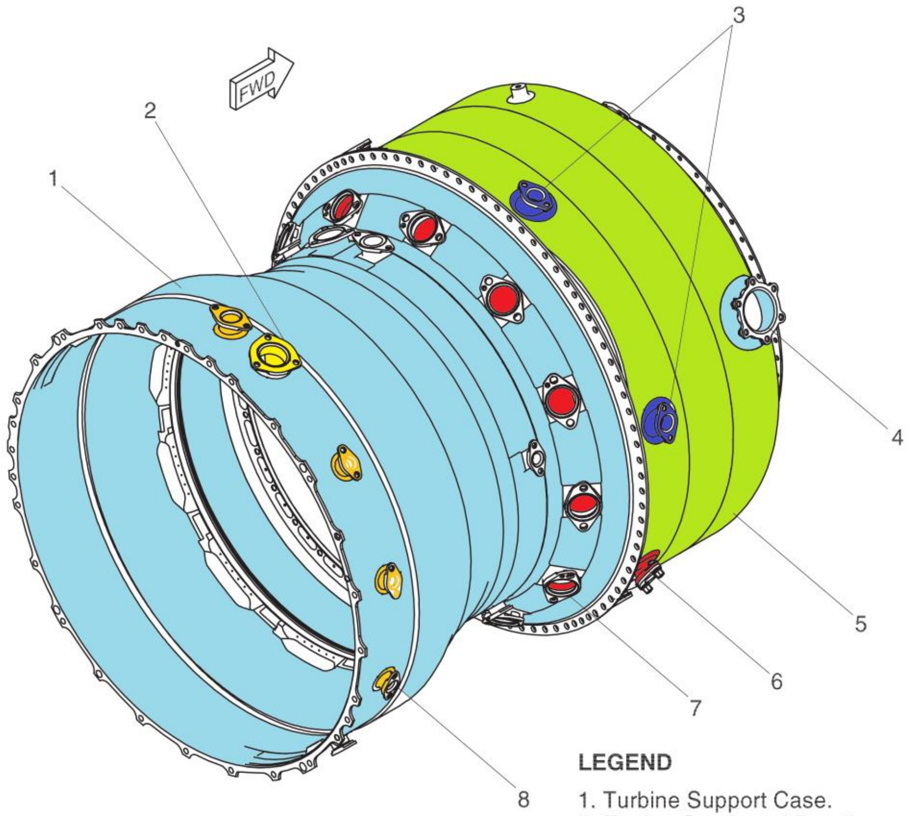
The inner surface comprises machined louvers. Two igniter plug bosses are provided on the gas generator case at the four and seven O'clock positions, with corresponding boss in the liner.

Combustor retention is provided by six combustor retention pins located in the igniter plane.

The major components comprising the combustor are the:

- Outer liner
- Inner liner
- Small exit duct (SED)
- Adapter.

The large exit duct is double skinned, with impingement holes in the outer skin for effective cooling of the inner skin.



LEGEND
1. Turbine Support Case.
2. Engine Overboard Breather.
3. Borescope Ports.
4. P3 Air Bleed Pad.
5. Gas Generator Case.
6. Igniter Plug Mounting Pad.
7. Fuel Nozzle Mounting Pad.
8. ITT Mounting Pads.

Figure 72-11 Combustion Section

High Pressure Combustor

Refer Figure 72-12 & *72-13*

Introduction

The combustor burns a mixture of fuel and air, and delivers the resulting gases to the turbines. Fuel is added to the compressed air in the combustion chamber. The igniters are used for starting but are not required once the fuel/air mixture is lit.

General Description

The annular reverse flow combustion chamber is contained in the gas generator case. It is located at the rear of the HP compressor and the fish tail diffuser ducts. The major components comprising the combustor are the:

- Outer liner
- Inner liner
- Small Exit Duct (SED)
- Adapter
- Piston Ring

Detailed Description

Compressed air from the HP diffuser ducts flows into the Gas Generator Case around the combustion liner. The air flows into the dilution holes and through holes in the machined louvers. Air is used to assist in the fuel atomization, for cooling the liner skin, and for combustion. The large exit duct is double skinned, with impingement holes in the outer skin for effective cooling of the inner skin. The inner surface comprises machined louvers. Hot combustion gases flow forward in a three dimensional torroid and are directed rear toward the HP Vane by the combustor outer liner and the Small Exit duct.

The combustor is made of INCO−625 material, with integral drilled louvers. A thermal barrier coating (plasma spray coating) is applied to all the inside surfaces of the combustor, except for the louvers immediately adjacent to the hole exits. A fuel manifold is mounted around the exterior of the Turbine Support Case. Twelve hybrid fuel nozzles each containing a primary airblast, and a secondary airblast with combustor air swirler, are incorporated.

Two igniter plug bosses are provided on the gas generator case at the four and seven O’clock positions, with corresponding bosses in the liner. Combustor retention is provided by six combustor retention pins located in the igniter plane. Four of these bosses also permit borescope inspection of the combustor.

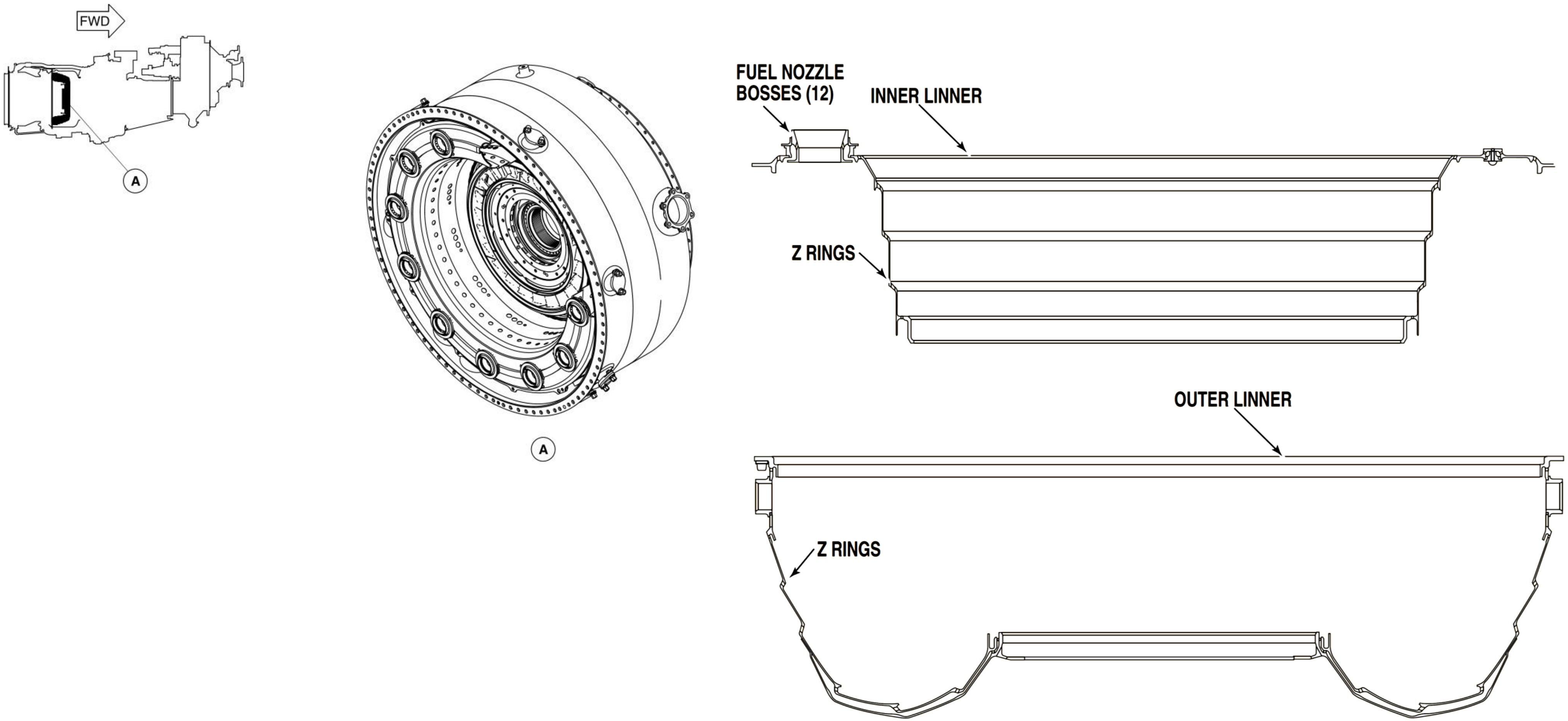
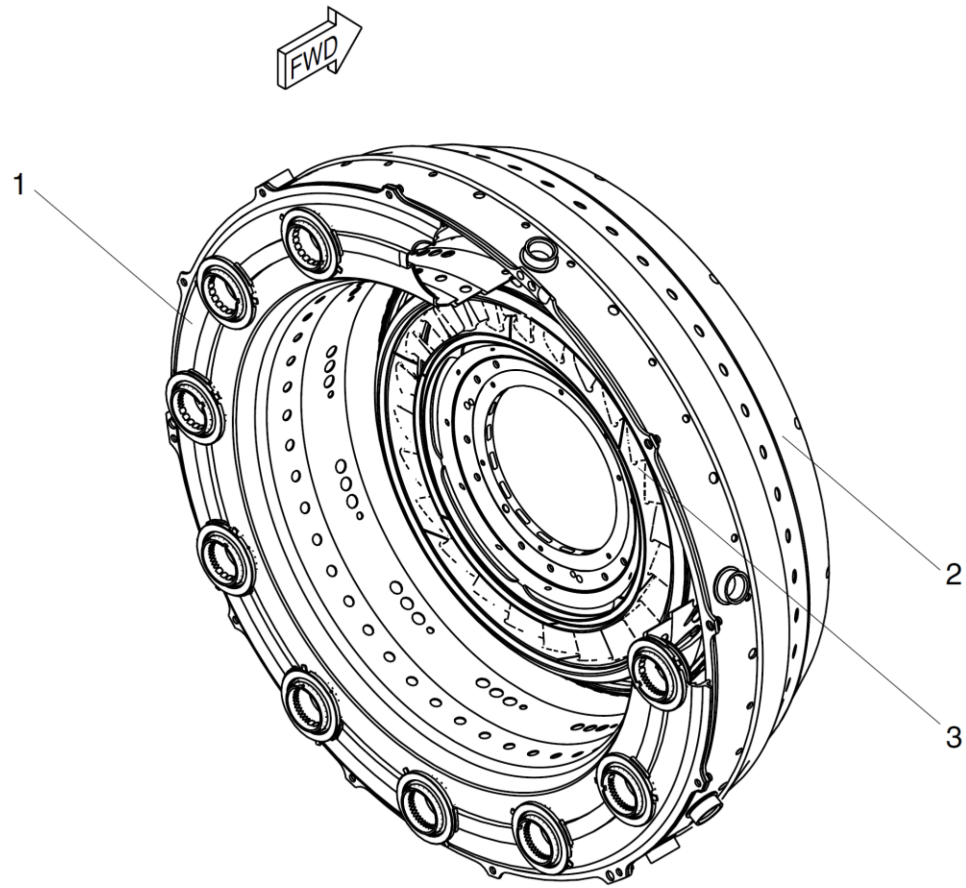


Figure 72-12 Combustion Section



LEGEND

1. Inner Liner.
2. Outer Liner.
3. H.P. Vane Assembly.

Figure 72-13 Combustion Chamber Liner

Turbines

Introduction

The turbines provide the power required to turn the propeller, compressor and engine accessories.

General Description

The turbines extract kinetic energy from the expanding gases as the gases come from the combustor. The turbines then convert this energy into shaft horsepower to drive the compressors, propeller and the engine accessories.

The hot section of the engine has the three turbine stages that follow:

- Low Pressure (LP) turbine
- High Pressure (HP) turbine
- Power Turbines (PT)

The purpose of the HP vane assembly is to direct the hot gases to the HP turbine and change static pressure into velocity. The vane stage is made of 28 airfoils, seven segments of four airfoils each. Support comes from the Small Exit Duct (SED) and the inner support housing. The vanes are air−cooled, using showerhead cooling holes.

Detailed Description

Refer Figures 72-14 & *72-15*

The LP and HP turbines are installed in the rear of the gas generator case, and the power turbines are installed in the turbine support case. Concentric shafts connect the two−stage power turbine to the gearbox and the single−stage LP and HP turbines to the compressors. The central PT shaft is supported by the No.1 (ball), No.2 (roller), No.6.5 (roller) and No.7 (roller) bearings.

The intermediate LP turbine shaft is supported by the No.2.5 (roller), No.3 (ball) and No.6 (roller) bearings.

The HP turbine shaft, that is integral with the impeller, is supported by the No.4 (ball) and No.5 (roller) bearings.

The hot section also has the vane assemblies that follow:

- HP vane assembly
- LP vane assembly
- Inter−turbine vane assembly
- Power turbine vane assembly

Each vane assembly is installed in front of its associated turbine, to direct the hot gases to the turbines and change static pressure into velocity.

All three turbines turn at independent speeds. At engine start, the starter/generator turns only the HP compressor and HP turbine. The HP turbine rotates clockwise (pilot view) to a maximum speed of 31150 rpm (100%). The LP turbine rotates counterclockwise to a maximum speed of 27000 rpm (100%). The Power turbine rotates clockwise to a maximum speed of 17501 rpm (100%).

Operation

The hot section of the engine comprises components downstream of the gas generator. Hot expanding gases leaving the combustion chamber are directed towards the HP turbine blades by the HP turbine ring. After the HP turbine the hot gases are directed towards the LP turbine blades by the LP turbine ring.

The gases then travel across the PT vane rings and impinge on the PT turbine blades.

The LP and HP turbines turn the compressors through their own respective shafts. The power turbines turn the propeller through the power turbine shaft and the reduction gearbox.

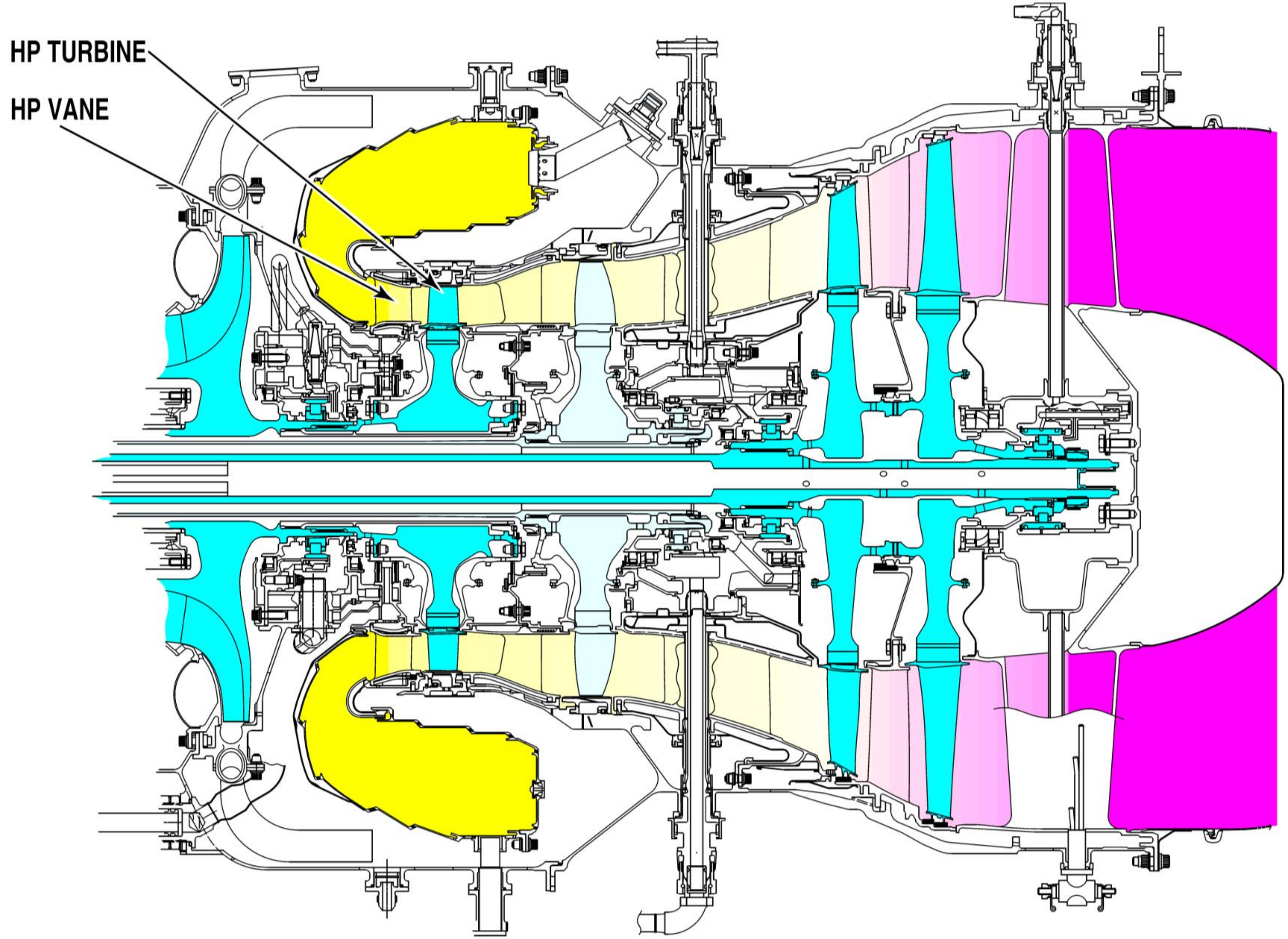


Figure 72-14 Turbines Locator – 1

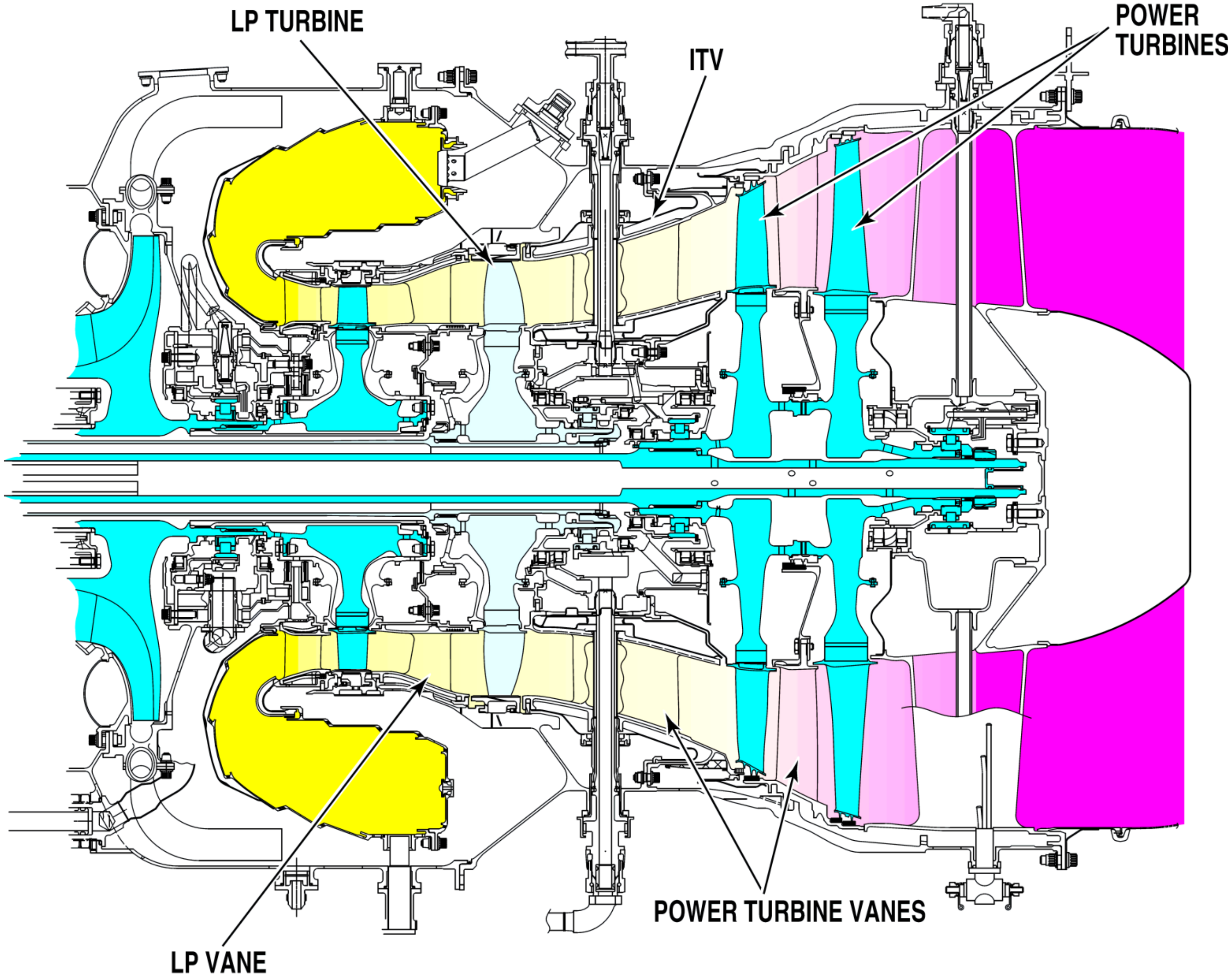


Figure 72-15 Turbines Locator – 2

Turbines – Component Description

High Pressure Turbine (HPT)

Refer Figure 72-16

The High Pressure Turbine consists of a nickel alloy disc featuring 41 air cooled blades. The base material for the blades is a single crystal nickel alloy. The blades are secured by fir−tree serrations and two covers. The cooling air is provided through showerhead, tip and platform cooling holes with trailing edge ejection. The HP disc transfers power from the expanding gases to the HP impeller. The disc has two covers that retain the blades axially and minimize cooling air leakage through the fir−tree of the blades. The front cover also increases the air pressure delivered. The cooling air comes from the Tangential Outboard Injector (TOBI). The cooling air is used to:

- Cool the HP blades
- Ventilate the HP disc bore
- Purge the downstream side of the rear cover
- Impinge upon the LP vane inner drum.

The disc is balanced by eccentric material removal. Both cover plates are detail balanced by concentric material removal. Assembly balancing is achieved by weights mounted on a scalloped flange.

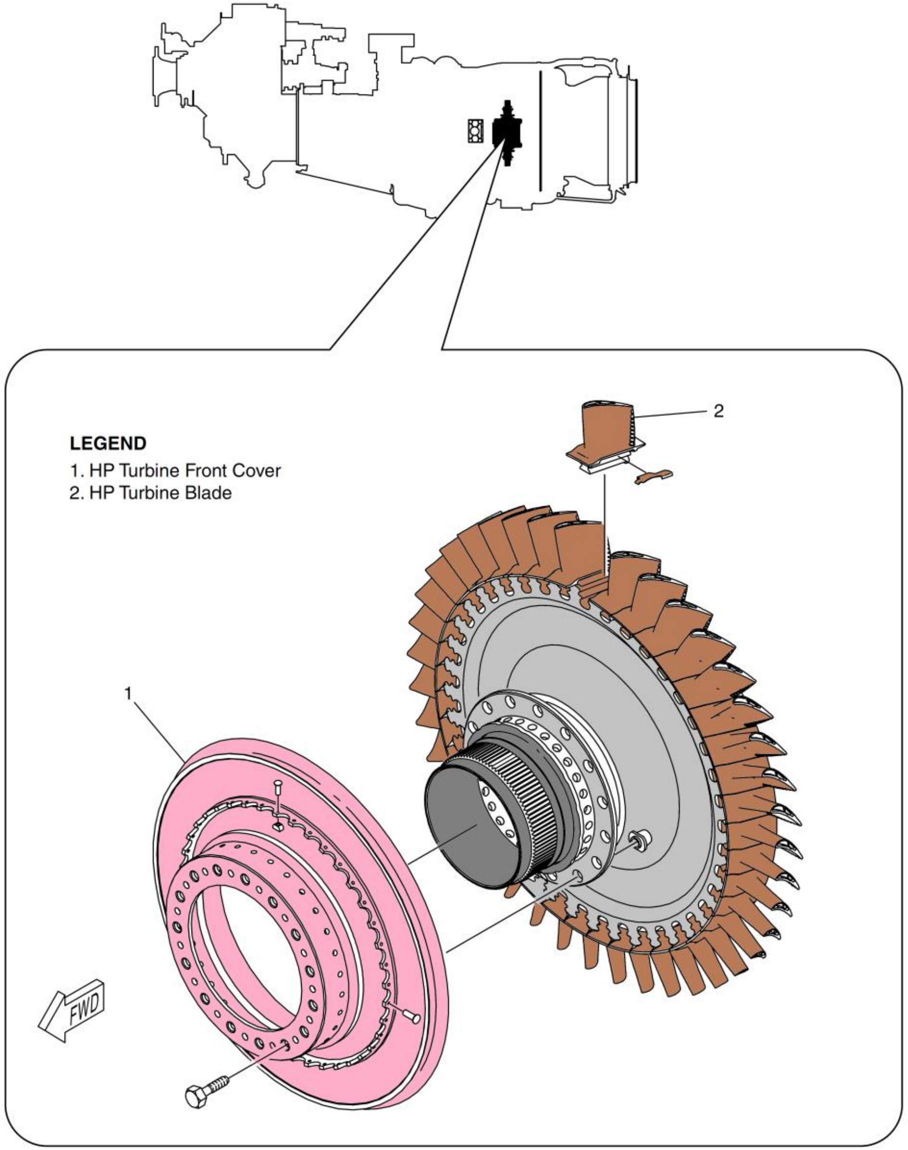


Figure 72-16 High Pressure Turbine

LP Turbine Vane Assembly

The purpose of the LP vane assembly is to direct the hot gases to the LP turbine and change static pressure into velocity.

The vane stage is made of 28 airfoils, seven segments of four airfoils each, assembled into a squirrel cage outer assembly. The outer support structure is made with large windows to facilitate impingement cooling on the vane segments and for weight saving. The squirrel cage also attenuates the distortion transmitted to the turbine support case to provide better tip clearance control.

LP Turbine

Refer Figure 72-17

The LP Turbine is made of nickel alloy. It consists of a disc featuring 41 air−cooled blades. The base material for the blades is a single crystal nickel alloy. The blades are secured by fir−tree serrations and integral rear face disc lugs with a blade retaining cover on the front face. The retaining cover is made with internal fins to pump air for the cooling of the blades. The blades use trailing edge ejection for the cooling air. The disc is straddle mounted on the LP shaft.

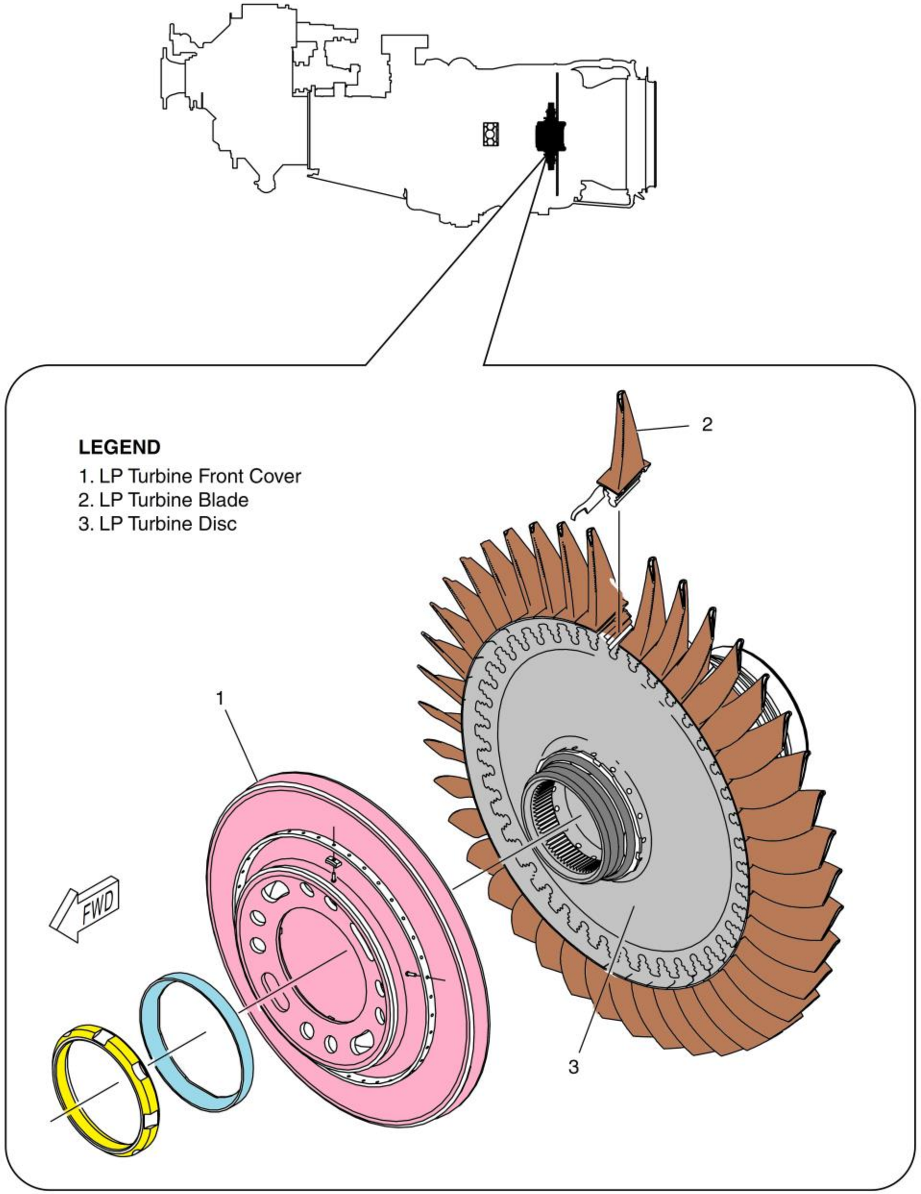


Figure 72-17 Low Pressure Turbine

Inter Turbine Vane (ITV) Assembly

The purpose of the LP vane assembly is to direct the hot gases to the LP turbine and change static pressure into velocity.

The ITV is attached to the TSC at a flange. The ITV performs the following functions:

- Acts as the inter turbine duct
- Acts as the PT1 vane
- Provides support to the No. 6 and No. 6.5 bearing Housings

The ITV is made of 16 cambered airfoils: two airfoils are thick for oil passage, and 14 airfoils are thin. Cooling flow comes from the intercompressor case (P2.7) through an external pipe. The outer drum is cooled by an impingement shield. The air is then passed through the airfoils using a multiple serpentine cooling arrangement. In the event of a LP shaft failure, The ITV is designed to contain the LP rotor axially.

Four borescope ports are part of the ITV for gas path component inspection.

Power Turbines

Refer Figure 72-18

The purpose the power turbines is to extract energy from the hot gases to turn the propeller through a reduction gearbox. The turbines consist of discs with blades. These make up rotor assemblies. A vane assembly is mounted between the rotors.

The rotor assemblies consist of 72 uncooled, shrouded blades secured to a disc through fir−tree serrations and rivets. The first stage blades are single crystal Nickel alloy. The second stage blades are Inconel alloy.

The Power Turbines Discs are made of Nickel alloy. A face coupling joint connects the discs. Each of the disc extension arm has a machined integral face coupling. Each member of the joint has 36 teeth, allowing for multiple indexing positions, should the need arise. This minimizes disc to disc run out.

The PT1 disc radial location is aft of the No. 6.5 bearing at two spigot locations. Similarly, the PT2 disc location is forward of No. 7 bearing, at two spigot locations. Each disc is installed and removed separately. This eases assembly and disassembly of the power turbine section.

The PT2 Vane Assembly comprises 66 hollow airfoils. It is located in the turbine support case, by 11 equally spaced dogs and slots. The inner drum has 11 slots. Cooling is used over the blade shroud to minimize PT1 blade tip clearance impingement.

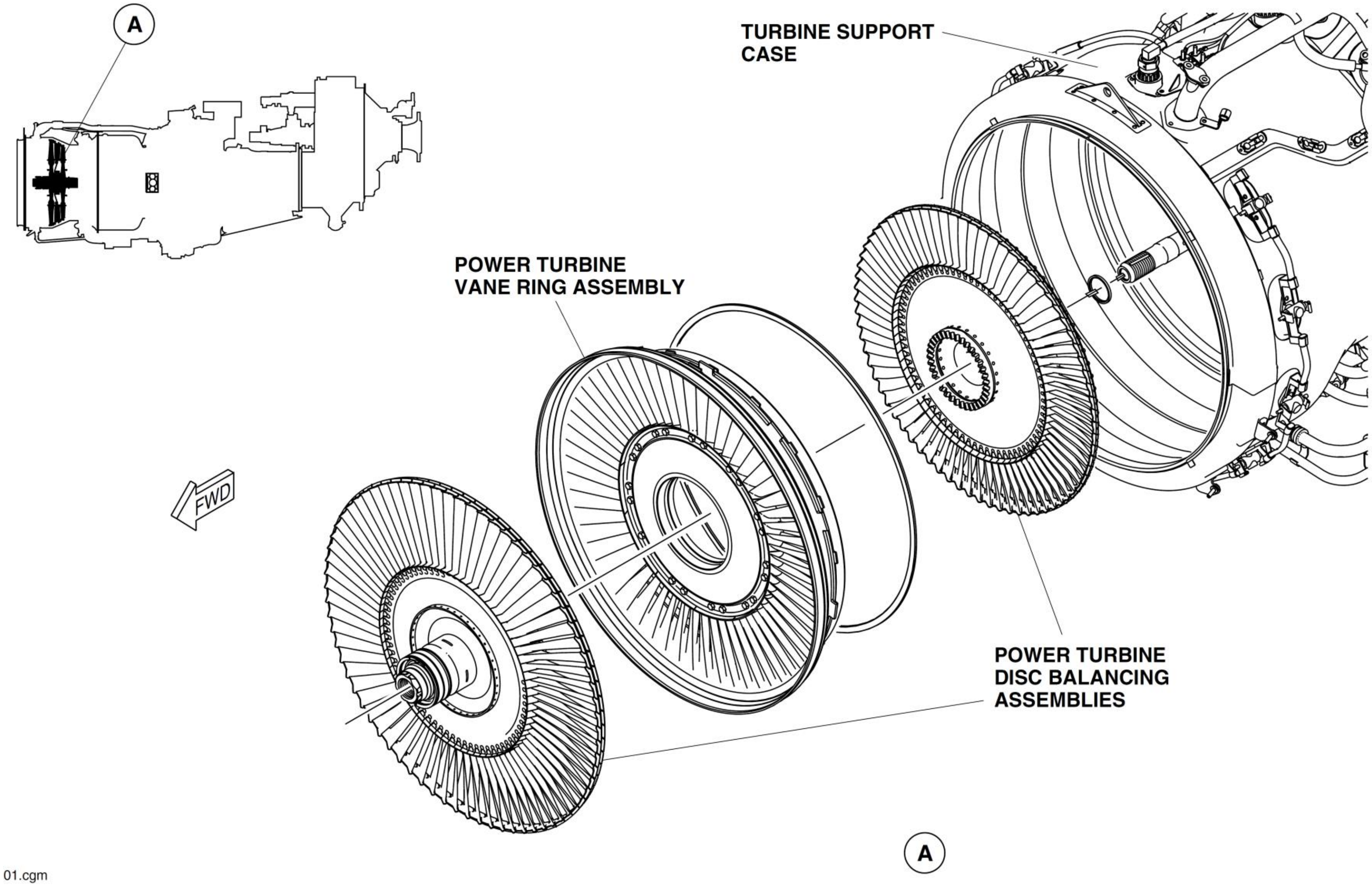


Figure 72-18 Power Turbines